original-19 / reliable

original-19 / delayed-spoof

original-19 / heavy-spoof

original-19 / reactive-spoof

synthetic-39 / reliable

synthetic-39 / delayed-spoof

synthetic-39 / heavy-spoof

synthetic-39 / reactive-spoof

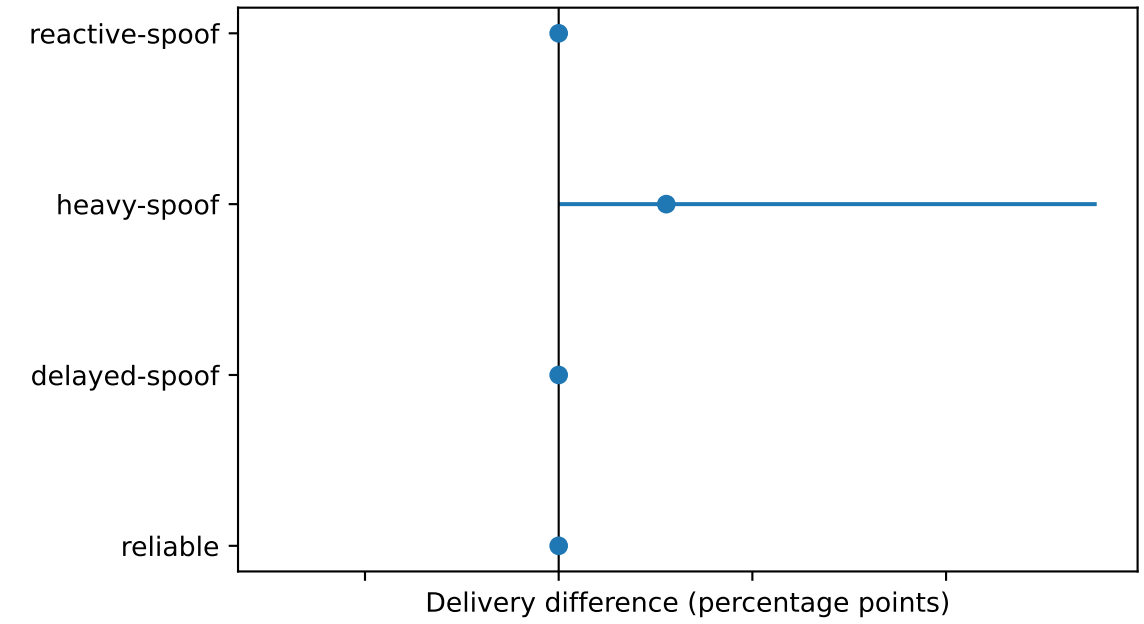
synthetic-51 / reliable

synthetic-51 / delayed-spoof

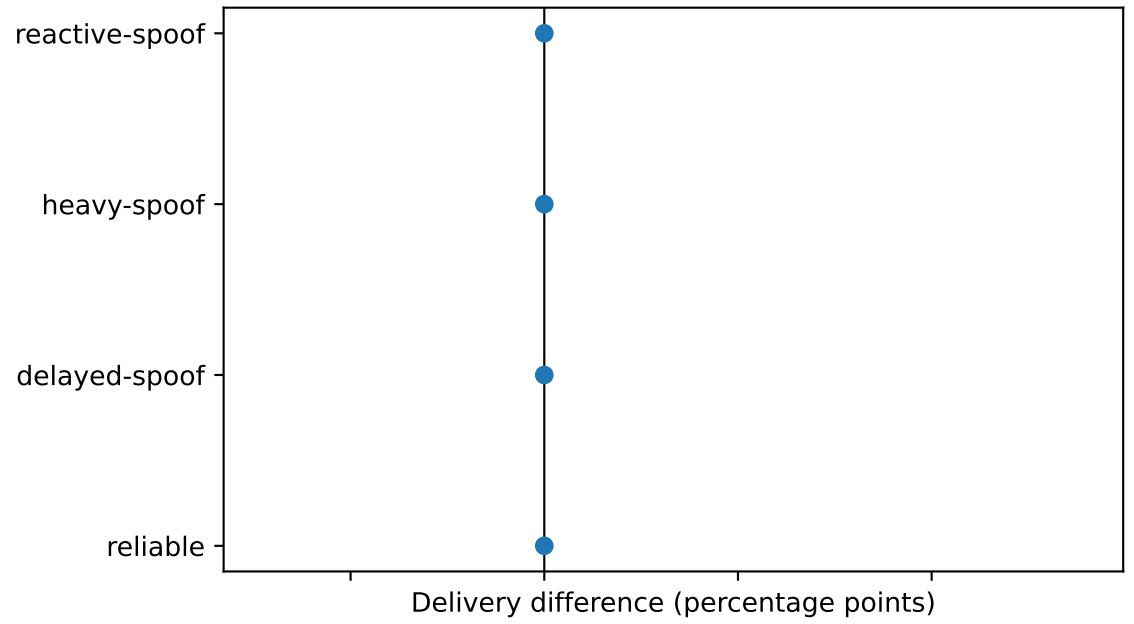
synthetic-51 / heavy-spoof

synthetic-51 / reactive-spoof

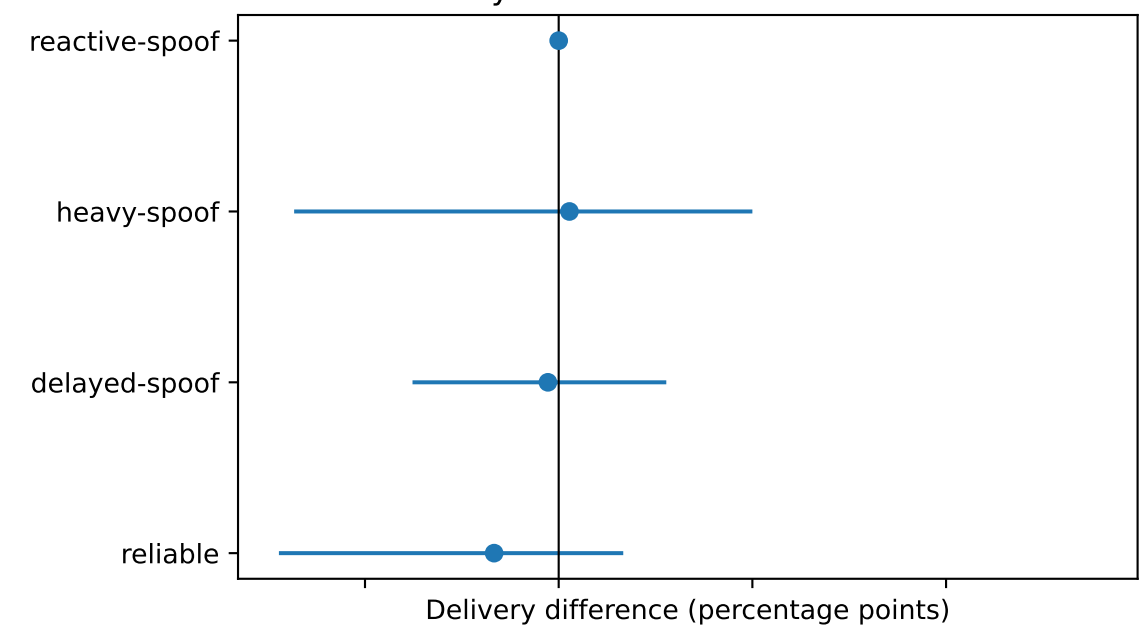
original-19 vs COMM-ALL



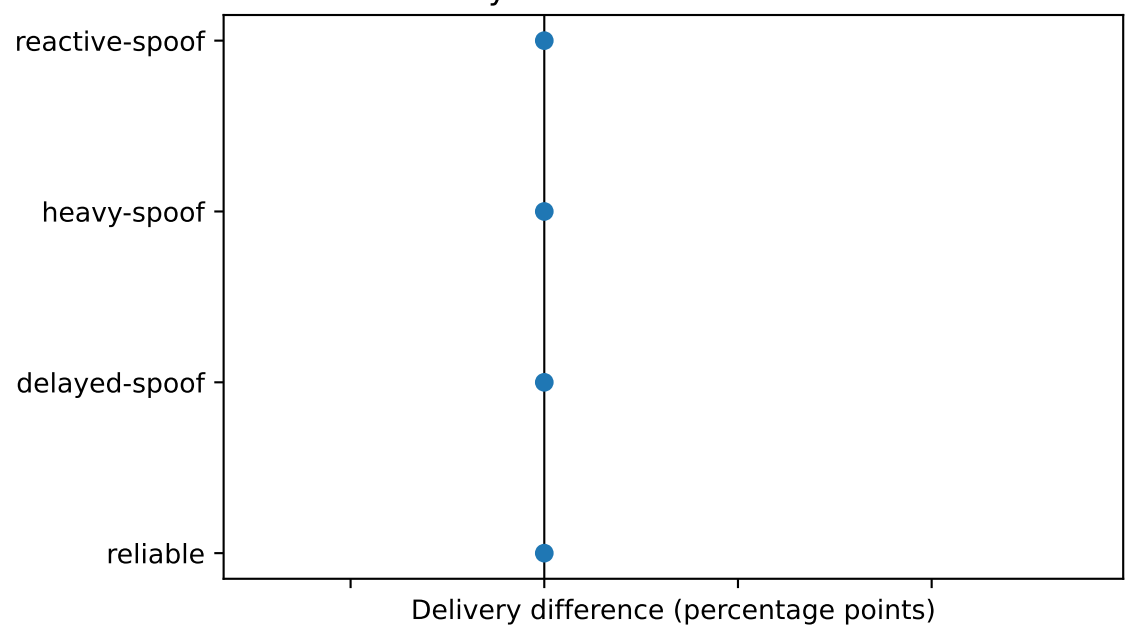
original-19 vs SOFT-ALL



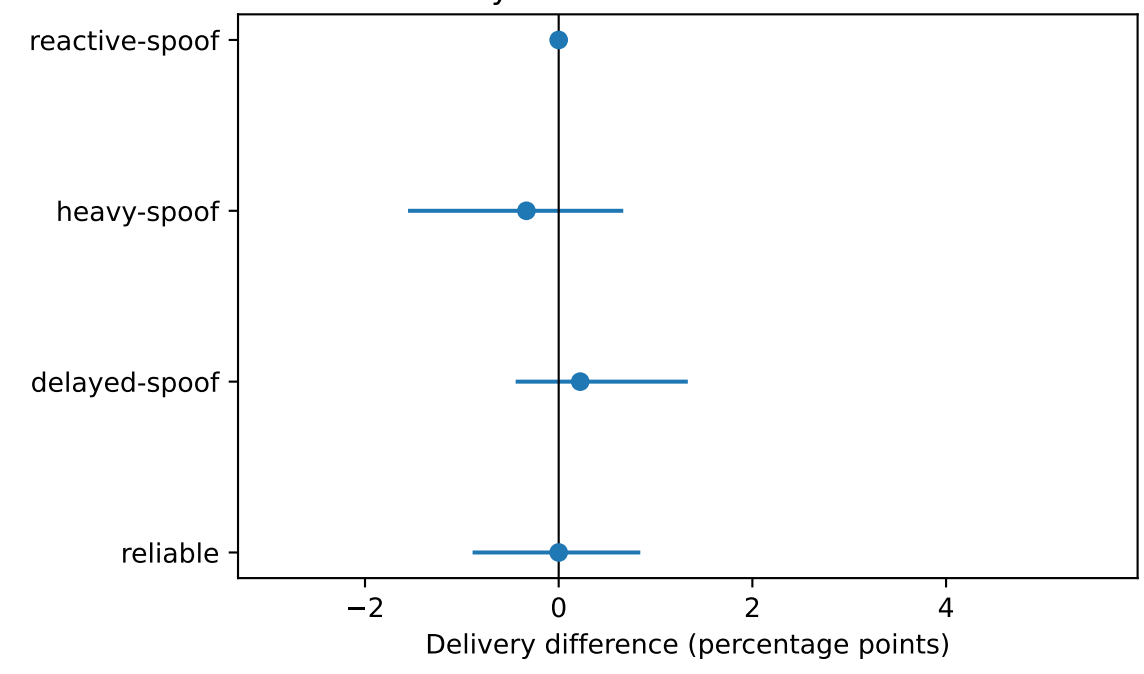
synthetic-39 vs COMM-ALL



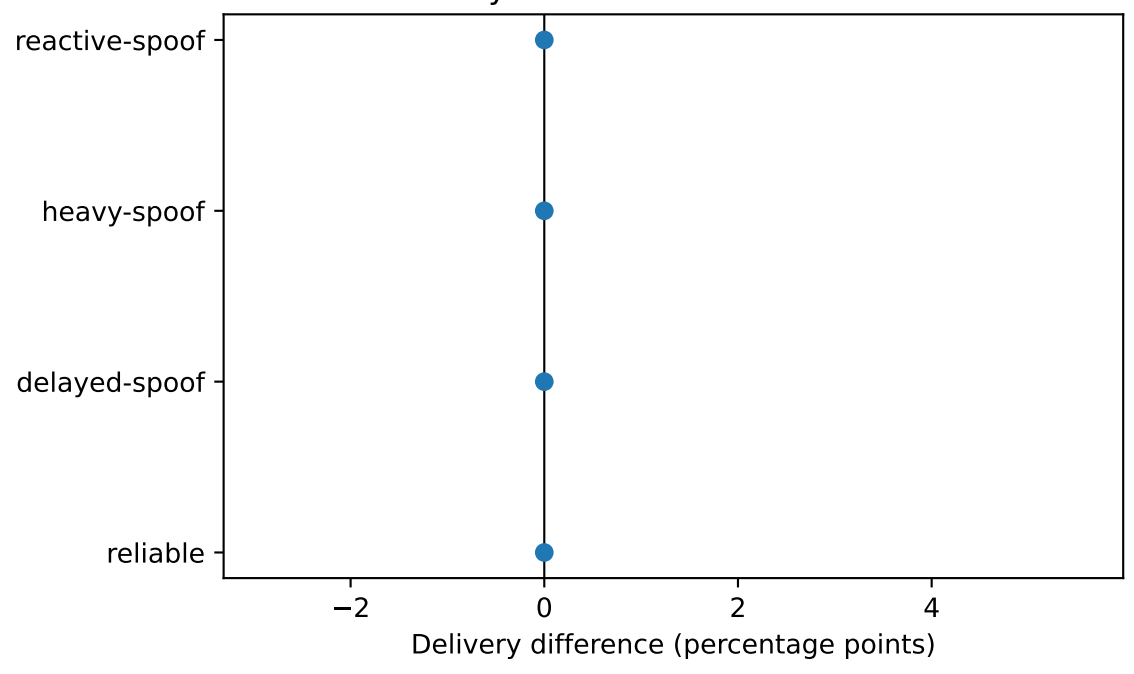
synthetic-39 vs SOFT-ALL

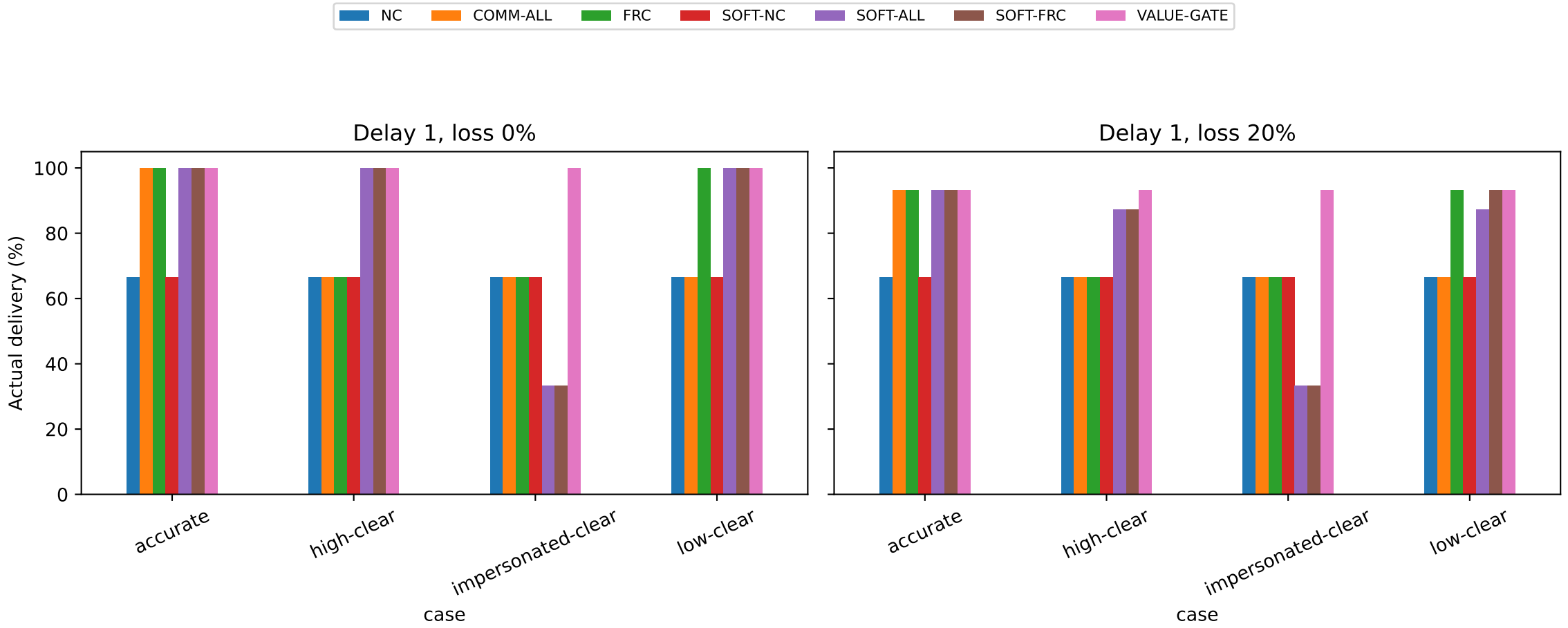


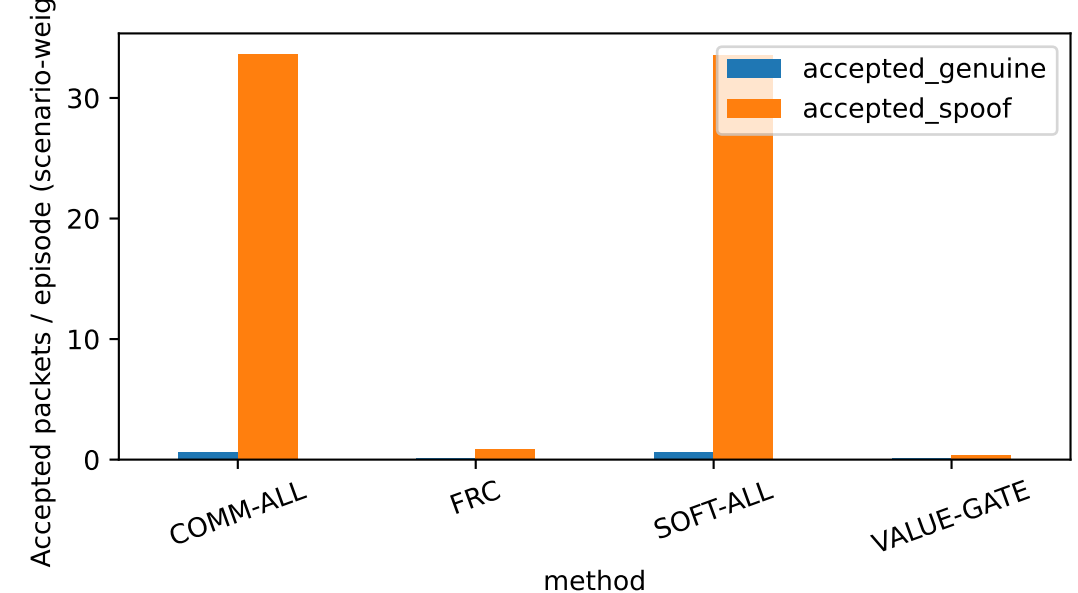
synthetic-51 vs COMM-ALL

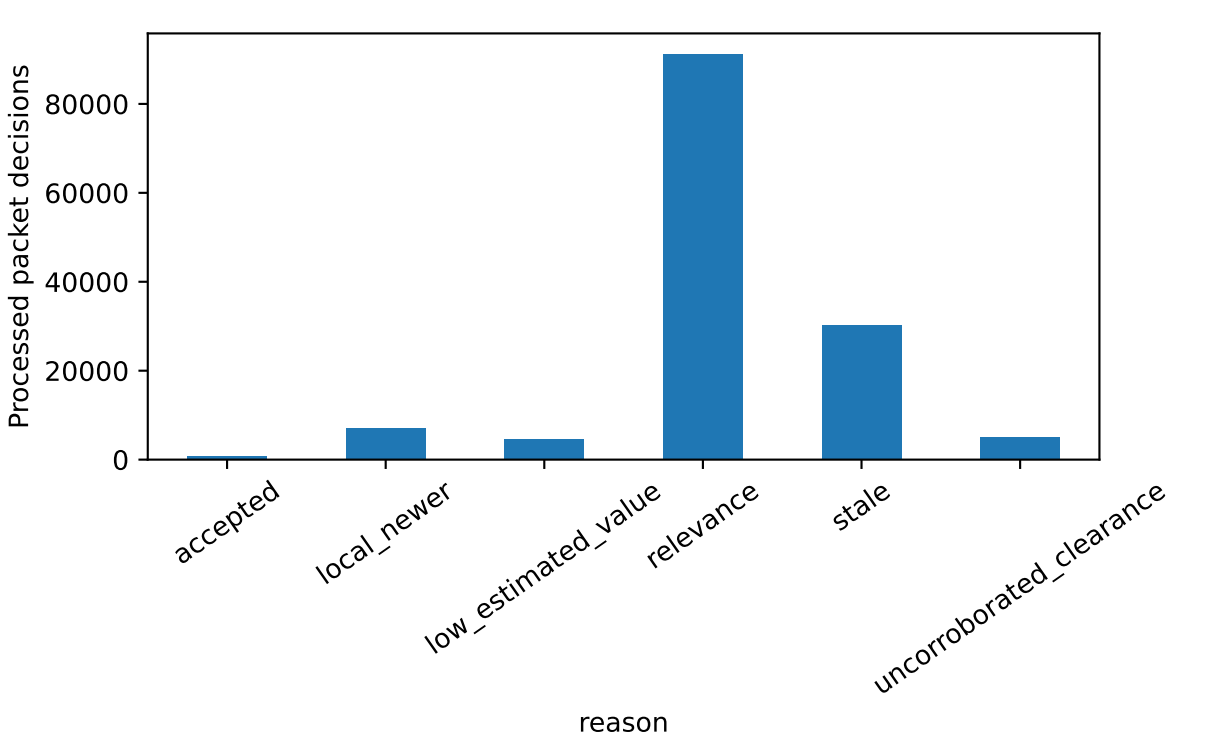


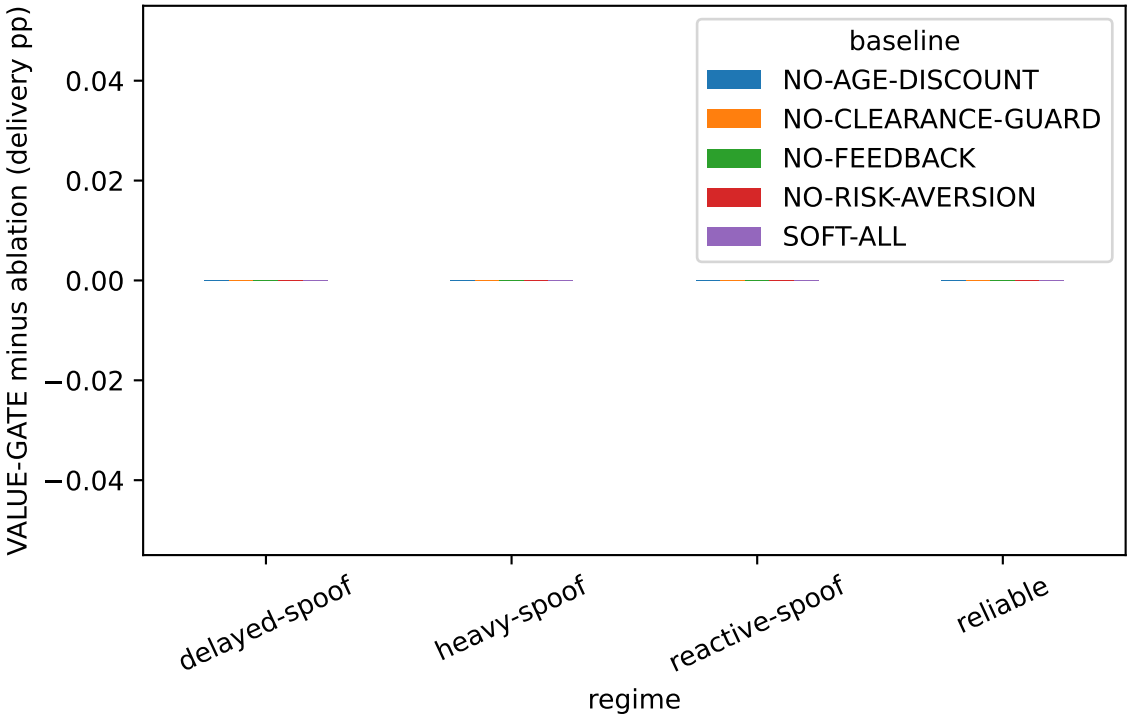
synthetic-51 vs SOFT-ALL

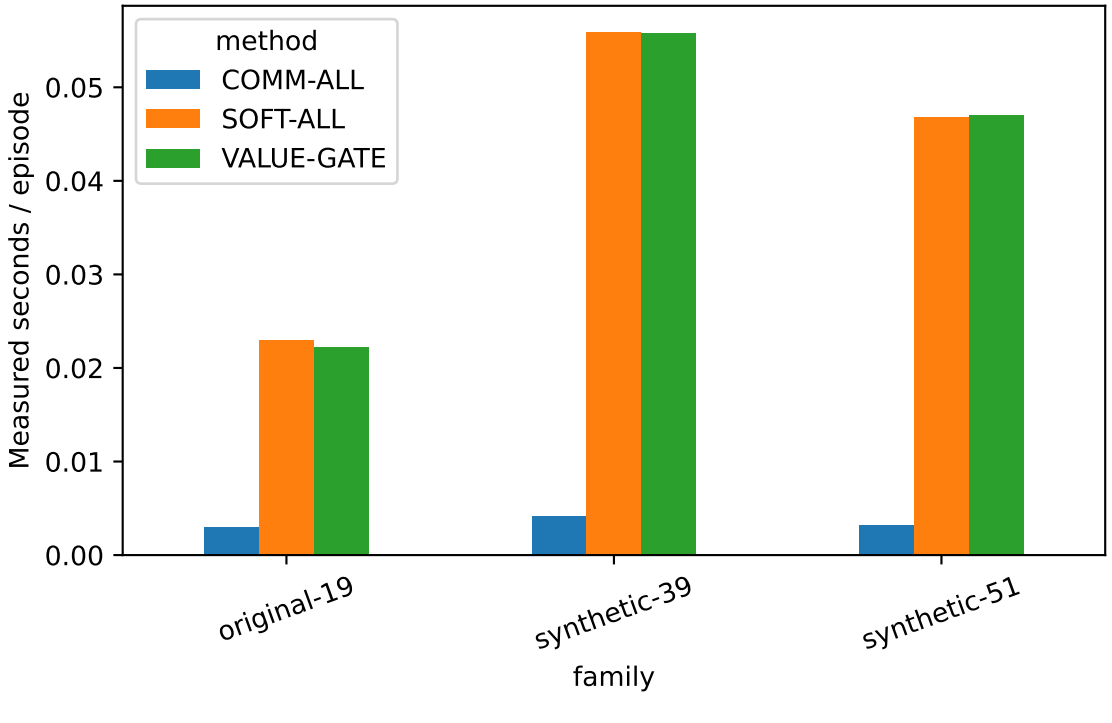


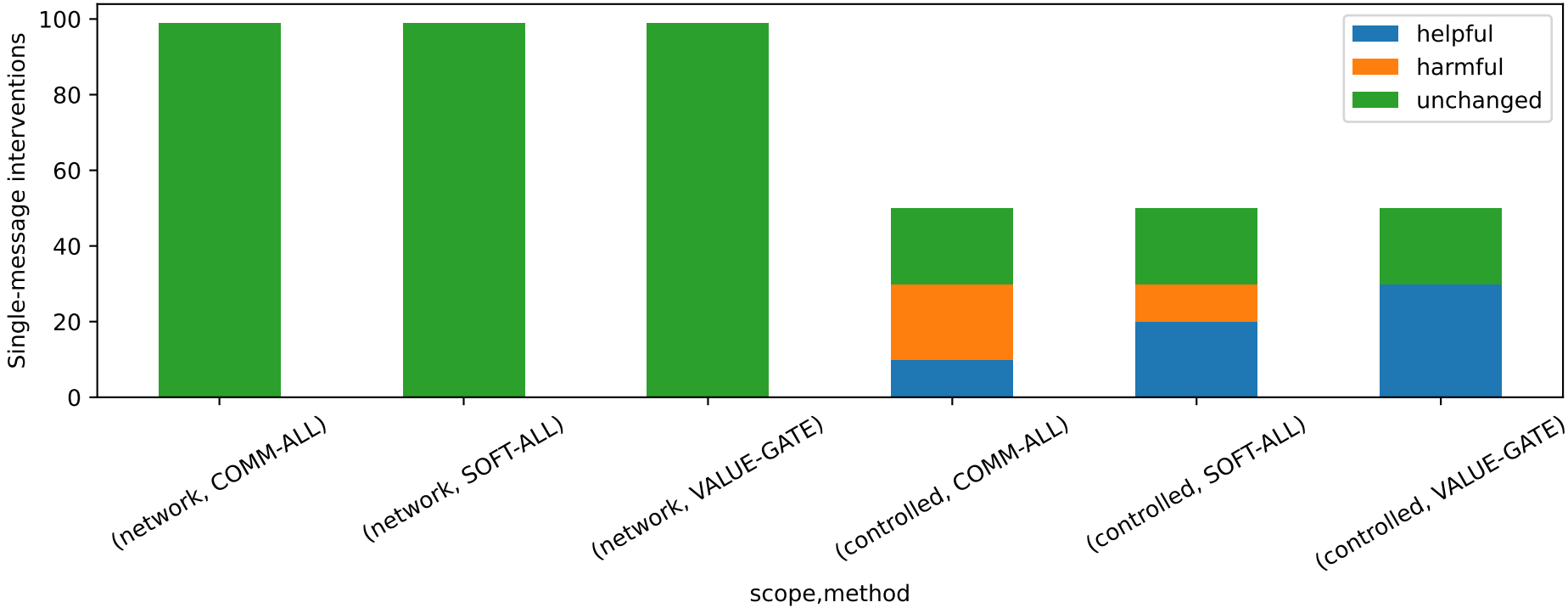

NC
COMM-ALL
FRC
SOFT-NC
SOFT-ALL
SOFT-FRC
VALUE-GATE
Delay 1, loss 0%
Delay 1, loss 20%
Actual delivery (%)
0
20
40
60
80
100
accurate
high-clear
impersonated-clear
low-clear
case

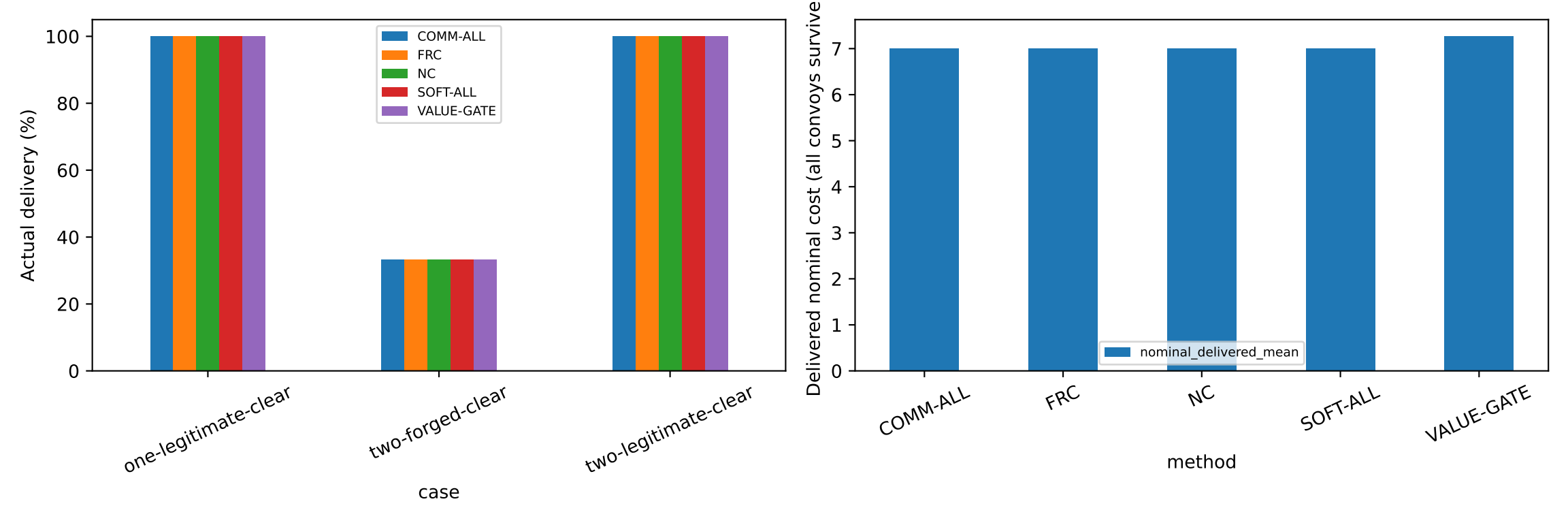

Accepted packets / episode (scenario-weig
0
10
20
30
accepted_genuine
accepted_spoof
COMM-ALL
FRC
SOFT-ALL
VALUE-GATE
method

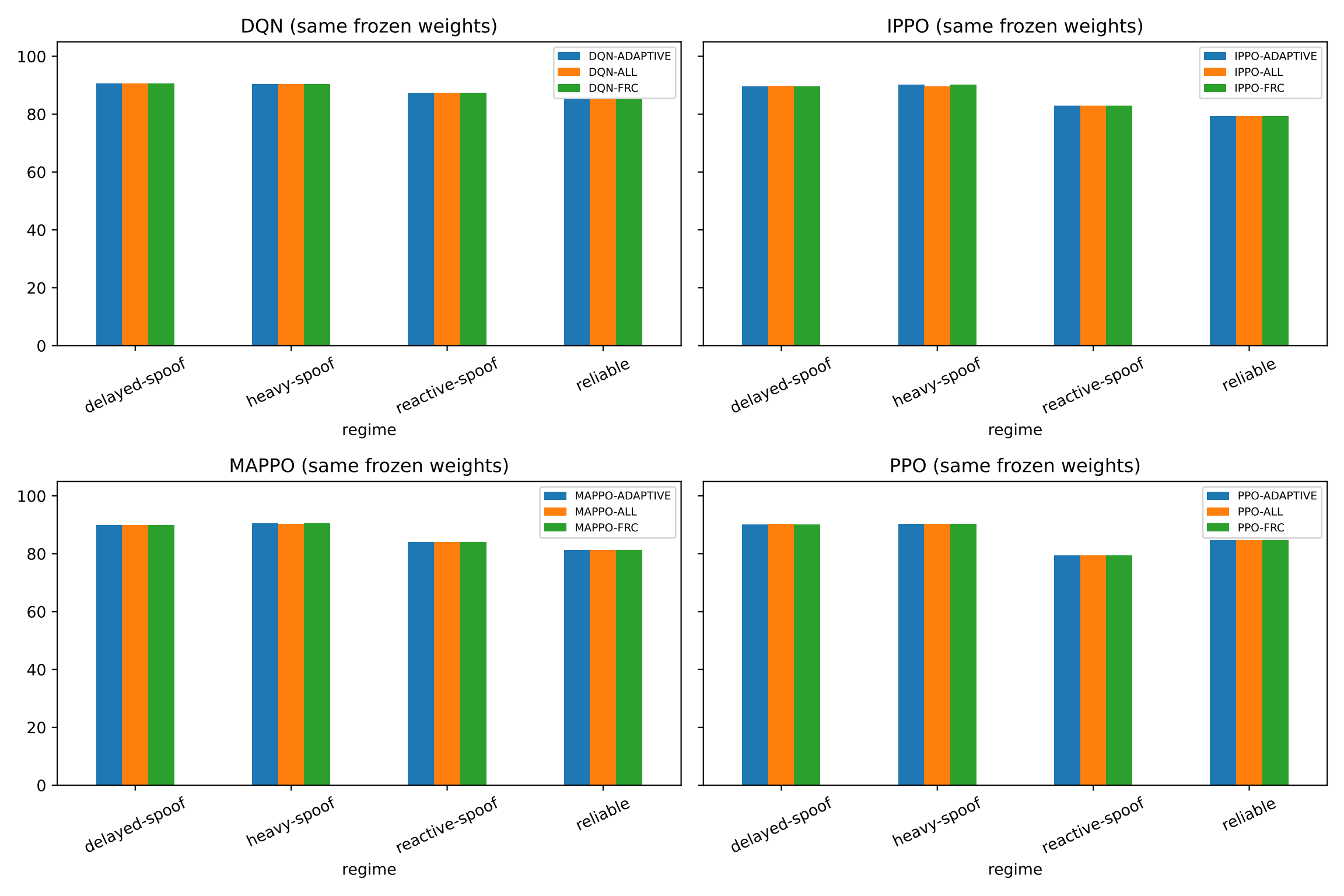

Processed packet decisions
0
20000
40000
60000
80000
accepted
local_newer
low_estimated_value
relevance
stale
uncorroborated_clearance
reason


VALUE-GATE minus ablation (delivery pp)
0.04
0.02
0.00
−0.02
−0.04
baseline
NO-AGE-DISCOUNT
NO-CLEARANCE-GUARD
NO-FEEDBACK
NO-RISK-AVERSION
SOFT-ALL
delayed-spoof
heavy-spoof
reactive-spoof
reliable
regime


Measured seconds / episode
0.05
0.04
0.03
0.02
0.01
0.00
method
COMM-ALL
SOFT-ALL
VALUE-GATE
original-19
synthetic-39
synthetic-51
family


Single-message interventions
100
80
60
40
20
0
helpful
harmful
unchanged
(network, COMM-ALL)
(network, SOFT-ALL)
(network, VALUE-GATE)
(controlled, COMM-ALL)
(controlled, SOFT-ALL)
(controlled, VALUE-GATE)
scope,method


Actual delivery (%)
100
80
60
40
20
0
COMM-ALL
FRC
NC
SOFT-ALL
VALUE-GATE
one-legitimate-clear
two-forged-clear
two-legitimate-clear
case
Delivered nominal cost (all convoys survive
7
6
5
4
3
2
1
0
nominal_delivered_mean
COMM-ALL
FRC
NC
SOFT-ALL
VALUE-GATE
method


DQN (same frozen weights)
DQN-ADAPTIVE
DQN-ALL
DQN-FRC
IPPO (same frozen weights)
IPPO-ADAPTIVE
IPPO-ALL
IPPO-FRC
MAPPO (same frozen weights)
MAPPO-ADAPTIVE
MAPPO-ALL
MAPPO-FRC
PPO (same frozen weights)
PPO-ADAPTIVE
PPO-ALL
PPO-FRC
0
20
40
60
80
100
delayed-spoof
heavy-spoof
reactive-spoof
reliable
regime